

Nils Lukas

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Research Interests

Design secure and private Machine Learning systems in the presence of untrustworthy

1. **Providers:** Confidential computing via Homomorphic Encryption & Secret Sharing.
2. **Data:** Mitigate data poisoning during training & prompt injection during inference.
3. **Models:** Protect training data privacy through PII scrubbing & differential privacy.
4. **Users:** Control misuse by detecting generated (mis)information with watermarking.

Education

University of Waterloo, Canada 2019 – 02/2024
Ph.D. in Computer Science

- Advisor: [Florian Kerschbaum](#)
- Thesis: Analyzing Threats of Large-Scale Machine Learning Systems
- Thesis Awards: [Top Mathematics Doctoral Prize](#) & [Alumni Gold Medal](#)

RWTH-Aachen, Germany
M.Sc. in Computer Science (*w/Distinction*) 2016 – 2018
B.Sc. in Computer Science 10/2012 – 2016

Honors & Awards

[Amazon Research Award](#) [100,000 USD] 2025
First Place at the NeurIPS'24 Watermarking Competition [4,400 USD] 2024
First Place at DGE Elite Hackathon, GITEX'24 [10,900 USD] 2024
Top Mathematics Doctoral Thesis, University of Waterloo [1,080 USD] 2024
Alumni Gold Medal, One PhD Award Yearly, University of Waterloo 2024
Best Poster Award, Sponsored by David R. Cheriton [220 USD] 2023
Distinguished Contribution Award, Microsoft MLADS conference 2023
David R. Cheriton Scholarship, University of Waterloo [14,400 USD] 2022, 2023
Outstanding Reviewer, ICML'22 2022
Best Poster Award, Sponsored by Rogers [720 USD] 2019

Research Grants

Awarded **\$1,322,000**
[IAAI Research Program] IAAI Knowledge Pipeline **\$500,000** · 2026
PI: Salem Lahlou · Co-PIs: Nils Lukas, Martin Takáč
[Amazon Research Awards – Special Call] AdvSim2Real: Robustifying Agents in Simulated Adversarial Environments **\$100,000** · 2025
PI: Nils Lukas · Co-PI: [Praneeth Vepakomma](#)
[Etihad Airways] Conversational Booking Agents **\$450,000** · 2025
PI: Nils Lukas · Co-PIs: Salem Lahlou, Alham Fikri, Martin Takáč, Mingming Gong
[United Al-Saqer Group] Privacy-preserving Brain-Computer Interfaces **\$136,000** · 2025
PI: [Abdulrahman Mahmoud](#) · Co-PIs: Nils Lukas, [Elizabeth Churchill](#)
[TII Funding] GFlowNets for Fuzzing of Agentic Applications **\$136,000** · 2025
PIs: Salem Lahlou & Nils Lukas

Conference Publications

23
[ECCV'26] [SPQR: A Multi-Dimensional Benchmark for Safety Alignment under Benign Model Adaptation](#)
Mohammed Talha Alam, Nada Saadi, Fahad Shamshad, Nils Lukas, Karthik Nandakumar, Fakhri Karray, Samuele Poppi. European Conference on Computer Vision, 2026.

22
[CVPR'26] [RAVEN: Erasing Invisible Watermarks via Novel View Synthesis](#)
Fahad Shamshad, Nils Lukas, Karthik Nandakumar. IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026.

🏆 Oral (top 0.8%)

21	Collaborative Threshold Watermarking
[ICML'26]	Tameem Bakr, Anish Ambreth, Nils Lukas . The Forty-Third International Conference on Machine Learning, 2026.
20	CoRe: Collaborative Reasoning via Cross Teaching
[ICML'26]	Kshitij Mishra, Mirat Aubakirov, Martin Takáč, Nils Lukas , Salem Lahlou. The Forty-Third International Conference on Machine Learning, 2026.
19	Forest Before Trees: Latent Superposition for Efficient Visual Reasoning
[ACL'26] Main Conference	Yubo Wang, Juntian Zhang, Yichen Wu, Yankai Lin, Nils Lukas , Yuhan Liu. The 64th Annual Meeting of the Association for Computational Linguistics, 2026.
18	Detecting AI-Generated Video: A Vision-Language Dual-View Survey
[ACL'26] Findings	Dylan Xinming Hou, Juntian Zhang, Xu Gu, Yichen Wu, Nils Lukas , Gus Xia, Xiuying Chen, Yuhan Liu. Findings of the Association for Computational Linguistics, 2026.
17	DP-Fusion: Token-Level Differentially Private Inference for Large Language Models
[ICLR'26]	Rushil Thareja, Preslav Nakov, Praneeth Vepakomma , Nils Lukas . The Fourteenth International Conference on Learning Representations, 2026.
16	SD-E2: Semantic Exploration for Reasoning Under Token Budgets
[EACL'26] Findings	Kshitij Mishra, Nils Lukas , Salem Lahlou. Findings of the European Chapter of the Association for Computational Linguistics, 2026.
15	Mask Image Watermarking
[NeurIPS'25] AR: 24.5% (5,290/21,575)	Runyi Hu, Jie Zhang, Shiqian Zhao, Nils Lukas , Jiwei Li, Qing Guo, Han Qiu, Tianwei Zhang. The Thirty-Ninth Annual Conference on Neural Information Processing Systems, 2025.
14	SPIRIT: Patching Speech Language Models against Jailbreak Attacks
[EMNLP'25] AR: 22.2% (1,811/8,172)	Amir Djanibekov, Nurdaulet Mukhituly, Kentaro Inui, Hanan Aldarmaki, Nils Lukas . Empirical Methods in Natural Language Processing (Main Conference), 2025.
13	Optimizing Adaptive Attacks against Content Watermarks for Language Models
[ICML'25] AR: 26.9% (3,260/12,107) ★ Spotlight (Top 2.6%)	Abdulrahman Diao, Toluwani Aremu, Nils Lukas . The Forty-Second International Conference on Machine Learning, 2025.
12	Cowpox: Towards the Immunity of VLM-based Multi-Agent Systems
[ICML'25] AR: 26.9% (3,260/12,107)	Yutong Wu, Jie Zhang, Yiming Li, Chao Zhang, Qing Guo, Han Qiu, Nils Lukas , Tianwei Zhang. The Forty-Second International Conference on Machine Learning, 2025.
11	PEPSI: Practically Efficient Private Set Intersection in the Unbalanced Setting
[USENIX'24] AR: 19.1% (417/2,176)	Rasoul Mahdavi, Nils Lukas , Faezeh Ebrahimiaghazani, Thomas Humphries, Bailey Kacsmar, John Premkumar, Xinda Li, Simon Oya, Ehsan Amjadian, Florian Kerschbaum. In the 33rd USENIX Security Symposium, 2024.

10	Fast and Private Inference of Deep Neural Networks by Co-designing Activation Functions
[USENIX'24]	Abdulrahman Diaa, Lucas Fenaux, Thomas Humphries, Marian Dietz, Faezeh Ebrahimiaghazani, Bailey Kacsmar, Xinda Li, Nils Lukas , Rasoul Akhavan Mahdavi, Simon Oya, Ehsan Amjadian, Florian Kerschbaum. In the 33rd USENIX Security Symposium, 2024.
AR: 19.1% (417/2,176)	
9	Leveraging Optimization for Adaptive Attacks on Image Watermarks
[ICLR'24]	Nils Lukas , Abdulrahman Diaa, Lucas Fenaux, Florian Kerschbaum. In the Twelfth International Conference on Learning Representations, 2024.
AR: 30.8% (2,250/7,262)	
8	Universal Backdoor Attacks
[ICLR'24]	Benjamin Schneider, Nils Lukas , Florian Kerschbaum. In the Twelfth International Conference on Learning Representations, 2024.
AR: 30.8% (2,250/7,262)	
● Media Coverage	
7	PTW: Pivotal Tuning Watermarking for Pre-Trained Image Generators
[USENIX'23]	Nils Lukas and Florian Kerschbaum. In the 32nd USENIX Security Symposium, 2023.
AR: 29.2% (422/1,444)	
6	Analyzing Leakage of Personally Identifiable Information in Language Models
[S&P'23]	Nils Lukas , Ahmed Salem, Robert Sim, Shruti Tople, Lukas Wutschitz, Santiago Zanella-Béguelin. In the 44th IEEE Symposium on Security and Privacy, 2023.
AR: 17.0% (195/1,147)	
★ Distinguished Contribution Award at Microsoft MLADS	
5	SoK: How Robust is Image Classification Deep Neural Network Watermarking?
[S&P'22]	Nils Lukas , Edward Jiang, Xinda Li, Florian Kerschbaum. In the 43rd IEEE Symposium on Security and Privacy, 2022.
AR: 14.5% (147/1,012)	
4	Deep Neural Network Fingerprinting by Conferrable Adversarial Examples
[ICLR'21]	Nils Lukas , Yuxuan Zhang, Florian Kerschbaum. The Ninth International Conference on Learning Representations, 2021.
AR: 28.7% (860/2,997)	
★ Spotlight (Top 5%)	
3	On the Robustness of Backdoor-based Watermarking in Deep Neural Networks
[IH&MMSEC'21]	Masoumeh Shafieinejad, Nils Lukas , Jiaqi Wang, Xinda Li, Florian Kerschbaum. Proceedings of the 2021 ACM Workshop on Information Hiding and Multimedia Security, 2021.
AR: 40.3% (128/318)	
2	Practical Over-Threshold Multi-Party Private Set Intersection
[ACSAC'20]	Rasoul Mahdavi, Thomas Humphries, Bailey Kacsmar, Simeon Krastnikov, Nils Lukas , John Premkumar, Masoumeh Shafieinejad, Simon Oya, Florian Kerschbaum, Erik-Oliver Blass. Annual Computer Security Applications Conference (ACSAC), 2020.
AR: 20.9% (104/497)	
1	Differentially Private Two-Party Set Operations
[EuroS&P'20]	Bailey Kacsmar, Basit Khurram, Nils Lukas , Alexander Norton, Masoumeh Shafieinejad, Zhiwei Shang, Yaser Baseri, Maryam Sepehri, Simon Oya, Florian Kerschbaum. IEEE European Symposium on Security and Privacy (EuroS&P), 2020.
AR: 20.9% (39/187)	
Journal Publications	
2	Privacy-Preserving Machine Learning [Cryptography]
[IEEE S&P'23]	Florian Kerschbaum, Nils Lukas . IEEE Security & Privacy, Volume 21, Issue 6, 2023.

	¹ [AIP'18]	SunFlower: A new Solar Tower Simulation Method for use in Field Layout Optimization Pascal Richter, Gregor Heiming, Nils Lukas , Martin Frank. AIP Conference Proceedings, Volume 2033, Issue 1, 2018.	
Workshop Papers	³ [GenAI4Health]	Sanitizing Medical Documents with Differential Privacy using Large Language Models Rushil Thareja, Gautam Gupta, Preslav Nakov, Praneeth Vepakomma, Nils Lukas . 2025.	
	² [WMARK'25]	First-Place Solution to NeurIPS 2024 Invisible Watermark Removal Challenge Fahad Shamshad, Tameem Bakr, Yahia Salaheldin Shaaban, Noor Hazim Hussein, Karthik Nandakumar and Nils Lukas . The 1st Workshop on GenAI Watermarking, 2025.	
	¹ [WMARK'25] ★ Oral Presentation	Optimizing Adaptive Attacks against Content Watermarks for Language Models Abdulrahman Diaa, Toluwani Aremu, Nils Lukas . The 1st Workshop on GenAI Watermarking, 2025.	
Working Papers		Mitigating Watermark Forgery in Generative Models via Randomized Key Selection , Submitted. Toluwani Aremu, Noor Hazim Hussein, Munachiso S Nwadike, Samuele Poppi, Jie Zhang, Karthik Nandakumar, Neil Zhenqiang Gong, Nils Lukas . 2025. Robust and Calibrated Detection of Authentic Multimedia Content , Submitted. Sarim Hashmi, Abdelrahman Elsayed, Mohammed Talha Alam, Samuele Poppi, Nils Lukas . 2025.	
Research Talks		Adaptively Robust and Forgery-Resistant Watermarking • Meta (FAIR), hosted by Hady Elsahar	2025
		Optimizing Adaptive Attacks against Content Watermarks • DeepMind, hosted by David Stutz	2024
		• University of California, Berkeley, hosted by Dawn Song	2024
		Analyzing Leakage of Personal Information in Language Models • Microsoft M365, hosted by Robert Sim	2024
		• Meta, hosted by Will Bullock	2023
		• MongoDB, hosted by Marilyn George and Archita Agarwal	2023
		How Reliable is Watermarking for Image Generators? • Google, hosted by Somesh Jha	2023
		• University of California, Berkeley, hosted by Dawn Song	2023
Keynotes		Aviation Future Week , hosted by Emirates, Dubai	2024
		Cyber Energy Leadership Forum , Abu Dhabi	2024
Work Experience		Assistant Professor , MBZUAI, Abu Dhabi, UAE	since 08/2024
		Research Intern , Royal Bank of Canada, Borealis AI, Toronto	2024
		• Vertical Federated Learning, hosted by Kevin Wilson	
		Research Intern , Microsoft Research, Cambridge, UK	2022
		• Privacy for Language Models, hosted by Shruti Tople & Lukas Wutschitz	
		Research Assistant , RWTH-Aachen, Aachen	2014 – 2018
		Student Researcher , DSA Daten- und Systemtechnik GmbH, Aachen	2016
		Software Engineer Intern , A.R. Bayer DSP Systeme GmbH, Düsseldorf	2012
Teaching		Instructor , MBZUAI, UAE	

Service	• ML8509: Collaborative Machine Learning (w/ S. Horvath)	2026
	• ML8502: Machine Learning Security (14 weeks)	2025
	• ML807: Federated Learning (7 weeks)	2025
	• ML818: Emerging Topics in Trustworthy Machine Learning (4 weeks)	2024
	Teaching Assistant, University of Waterloo, Canada	
	• CS458/658: Computer Security and Privacy	2020, 2021
	• CS246 – Object Oriented Programming	2021
	Co-Instructor, RWTH-Aachen, Germany	
	• Course: Data-driven Medicine	2018
	Area Chair	
	• International Conference on Learning Representations (ICLR)	2026
	• International Conference on Machine Learning (ICML)	2026
	Program Committee	
	• ACM Conference on Computer and Communications Security (CCS)	2025
	• IEEE Symposium on Security and Privacy (IEEE S&P)	2025, 2026
	• Recent Advances in Intrusion Detection (RAID)	2024
	Artifact Evaluation Committee	
	• The ACM Conference on Computer and Communications Security (CCS)	2023, 2024
	Reviewer	
	• NETYS	2025
	• ACM TheWebConf (WWW)	2025
	• International Conference on Learning Representations (ICLR)	2024, 2025
	• International World Wide Web Conference (TheWebConf)	2024
	• Recent Advances in Intrusion Detection (RAID)	2023
	• Neural Information Processing Systems (NeurIPS)	2022, 2023
	• International Conference on Machine Learning (ICML)	2022, 2025
	• The Conference on Information and Knowledge Management (CIKM)	2020
PostDocs, Students	Other	
	• Sub-Reviewer, Proceedings on Privacy Enhancing Technologies (PETS)	2021–2023
	• Session Chair, IEEE Symposium on Security and Privacy (S&P)	2023
	• Organization, Workshop on Semantic Web Solutions for Large-Scale Biomedical Data Analytics (SeWeBMeDA)	2018
	• Chair for the invited faculty talk program, International Symposium on Trustworthy Foundation Models at MBZUAI	2025
	• Faculty Search Committee, Machine Learning Department at MBZUAI	2025
	• Admission's Committee, MBZUAI Machine Learning Department	2025
	Student Board Member, Cybersecurity and Privacy Institute	2022–2024
	School Advisory Committee on Appointments Liaison, CrySP Lab	2022
	Postdoctoral Researchers	
	• Samuele Poppi <i>[primary]</i>	since 2025
	• Kshitij Mishra <i>[secondary]</i> (w/ Martin Takáč & Salem Lahlou)	since 2025
	• Dushyant Chauhan <i>[secondary]</i> (w/ Martin Takáč & Salem Lahlou)	since 2025
	• Yuhan Liu <i>[secondary]</i> (w/ Martin Takáč & Salem Lahlou)	since 2025
	PhD Students	
	• Fabian Zhang (ML)	since 2025
	• Tair Djanibekov (NLP)	since 2025
	• Rushil Thareja (NLP) <i>[primary]</i> (w/ Praneeth Vepakomma)	since 2024
	• Toluwani Aremu (ML)	since 2023

MSc Students

- Juan Nicolas Sepulveda Arias (CV) since 2025
- Mohamed Mahmoud Mohamed Hendy (ML) since 2025
- Majid Mohamed Mohamed Sulaiman Ibrahim (ML) since 2025
- Prince Jha (ML) *[secondary]* (w/ *Salem Lahlou*) since 2025
- Muhammad Mohideen (CV) (w/ *Samuel Horvath*) since 2025

Alumni

- Tameem Bakr (ML) MSc · Mar 2026
- Shaikha Jasem Mohamed Ismail Alhosani (ML) MSc · Mar 2026
- Maryam Abdulla Rashed Abdulla Alshamsi (ML) MSc · Mar 2026
- Maryam Alshehyari (ML) MSc · Mar 2026
- Abdalla Khalid Mohamed Abdalla Alzaabi (ML) MSc · Mar 2026
- Ali Al-Ali (ML) MSc · Mar 2026